
BRILLIANT METAL WORKS

Custom Metal Design & Fabrication

Project Management & Estimation System

USER GUIDE

Live Estimation

Section Optimizer

Margin Control

Rhino/GH Export

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1. System Overview

This system is your central tool for managing every railing and metalwork project — from the first client enquiry through to final installation. It removes guesswork from estimating, protects your margins, and produces professional client quotations in seconds.

What the system does

- Calculates material quantities — steel profiles, infill sheets, catch frames — automatically from dimensions
- Optimises panel sections — finds the cheapest combination of sheet size and number of sections
- Controls your pricing — adds consumables, surface treatment, transport and a configurable margin
- Generates client quotations — clean PDF with high-level categories — no internal cost breakdown shown
- Exports to Rhino/Grasshopper — one click sends exact geometry data to your CAD model

The three-tab workflow

STEP 1 Site Survey	STEP 2 Estimation	STEP 3 Quotation
Enter dimensions, select profiles and infill type. The system calculates everything live as you type.	Review the detailed material and labour breakdown. Add variable costs and set your margin. See the full pricing chain.	Preview the client document, download the PDF. The client sees professional categories — not your internal costs.

Project status flow

Every project moves through eight stages. Each stage is one click — you control when to advance.

Inquiry	Site Survey	Design	Quote Sent	Approved	Fabrication	Installation	Completed
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 You can also jump directly to any status by clicking it in the stepper bar at the top of a project page.

2. Dashboard & Project Management

Creating a new project

From the dashboard, click **+ New Project**. Fill in:

- Project code — assigned automatically (e.g. BAL-001 for balconies, STA-001 for staircases)
- Client name and phone number
- Location of the site
- Project type — Balcony or Staircase

Dashboard — search and filter

The dashboard shows all projects in a single table. Two tools help you find what you need quickly:

- Search bar — type any part of the project code, client name, title or location
- Status filters — click any status badge to show only projects at that stage

The project count updates live as you filter.

Project types

Type	What it covers
Balcony	Horizontal railings — any length, wall or floor mounted, one or multiple sections
Staircase	Inclined railings along a stringer — rise, run, one or both sides, parallelogram panels

3. Site Survey — Balcony

Dimensions

Field	Unit	What to measure
Total Length	mm	Full run of the railing from end post to end post
Total Height	mm	Top of rail to floor/slab — typically 900–1100mm residential

Section Layout Optimizer

This is the most important feature of the survey tab. Instead of guessing how many sections to use, the system **calculates every possible option** (from 2 sections up to ~20) and ranks them by total material cost.

Sections	Section W	Cut Size	Sheet	Sheets	Total Cost
3	1940mm	1960×1020mm	2000×1000	3	RWF 42,000
4	1440mm	1460×1020mm	2000×1000 ×2	2	RWF 38,500
5	1140mm	1160×1020mm	2440×1220	5	RWF 51,000

Click any row to select that layout. The row turns black to confirm your choice. **The BEST badge** marks the cheapest option. A blue "×N" means multiple panels nest on one sheet — those options are often the most economical.

 Changing any dimension, profile or infill type automatically recalculates all options. You must re-select a layout after any change.

Steel Profiles

Profile	Size	Typical use
20×20 SHS	20 × 20mm	Inner catch frame (standard)
40×20 RHS	40 × 20mm	Bottom rail — slim horizontal
40×40 SHS	40 × 40mm	Posts and top rail — most common
60×40 RHS	60 × 40mm	Heavy-duty posts or top rail

Infill Types

- Laser-Cut Sheet — plain steel sheet — section optimizer active, catch frame required
- Glass Panels — toughened glass — manual post spacing, no catch frame
- Flat Bars — horizontal bar infill — manual post spacing, bar spacing input

Mounting & Site

- Wall-mounted — specify wall type (concrete, brick, block) — affects hardware cost recommendations
- Floor-mounted — base plate welded to floor slab
- Access Difficulty — Easy / Medium / Difficult — inform this from your site visit, affects labour estimate

4. Site Survey — Staircase

Staircase dimensions

For a staircase railing, you are **not building the stairs** — you are building the railing along them. Measure the structural envelope of the stair flight.

Field	Unit	What to measure
Total Rise	mm	Vertical height from ground to top landing — e.g. 2800mm
Total Run	mm	Horizontal distance from first step to last — e.g. 3500mm
Width	mm	Usable stair width — e.g. 1200mm
Handrail Height	mm	Height of top rail above stringer line — typically 1000mm
Number of Flights		1 = straight stair, 2 = with intermediate landing
Rail Sides		One side only, or both sides of the stair

 The system calculates stringer length automatically: $\sqrt{(\text{rise}^2 + \text{run}^2)}$. You do not need to measure along the slope.

Section optimizer — inclined panels

Staircase panels are **parallelograms**, not rectangles — they follow the slope of the stringer. The optimizer works along the stringer length and calculates:

- Section slope spacing — distance between posts measured along the slope
- Horizontal span — the structural span between posts — flagged if it exceeds 2000mm
- Cut slope edge × cut height — exact parallelogram dimensions for the laser/CNC cutter
- Bounding rectangle — the sheet area needed to cut one panel — used for sheet nesting

 The bounding height is larger than the panel height because of the slope offset. The optimizer accounts for this automatically.

CNC Panel dimensions

After choosing a section layout, the Estimation tab shows a blue card with the exact cutting dimensions for all panels. All panels in one flight are identical — rotate alternate panels 180° when nesting to save material.

5. Estimation & Pricing

Material & Labour Breakdown

The top table on the Estimation tab shows every cost line with quantity, unit and unit price — this is your **internal view**. The client never sees this table.

Line item	What it covers
Steel Frame	Posts, top rail, bottom rail, catch frame — priced per kg
Infill Sheets / Glass / Bars	Panel material — priced per sheet or per m ²
CNC Cutting	Laser or plasma cut — priced per metre of cut length
Labour	Fabrication and welding days — priced per day

Variable Costs

Below the breakdown, the Variable Costs card covers the hidden costs that often cause underquoting:

Cost	Default	How it works
Design + Laser Cutting	Manual	Enter the actual cost from your cutting supplier
Installation	Manual	Labour cost for on-site fitting — varies by job
Welding Consumables	7% of frame	Auto-calculated — adjust the % slider if your usage differs
Surface Treatment	None	Select Powder Coat or Paint — enter rate per m ² ; cost auto-calculates from infill area
Transport & Delivery	Manual	Flat fee — enters your fuel and vehicle cost
Hardware & Fixings	Manual	Anchors, bolts, rawl plugs — flat fee per job

The Pricing Chain

This is the core of the margin system. Every number flows in one direction — there are no hidden overrides.

Materials + Labour subtotal <i>(from the breakdown table)</i>
+ Welding consumables + Surface treatment + Transport + Hardware + Design + Installation
= Your direct cost
+ Contingency (default 10%) <i>covers material price risk and rework</i>
= Adjusted cost
+ Business margin (default 30%) <i>overhead, equipment, profit</i>
= QUOTED PRICE ← this is what the client pays

i Both the contingency % and business margin % are editable inputs — you can adjust them per job. The defaults (10% + 30%) are set to protect you from the most common causes of underquoting.

Why 30% margin is not "too high"

A 30% business margin on top of direct + contingency costs covers:

- Workshop rent, electricity, equipment maintenance

- Owner and management time not billed as direct labour
- Guarantee and warranty call-backs
- Slow periods when direct costs still run

If your margin feels high, check whether these costs are already included in your labour rate. If they are not, 30% is the minimum safe level.

6. Quotation Generation

What the client sees vs what you see

The system keeps two views completely separate. Your internal cost data never appears in the client document.

Your internal Estimation tab	Client PDF & preview
• Each item with qty, unit, unit price • Frame cost, infill cost, labour cost • Consumables % and computed amount • Surface treatment rate per m² • Full pricing chain (direct → margin)	• Steel Frame & Fabrication • Infill Panels • Surface Treatment (if used) • Transport & Delivery (if used) • Installation (if used) <i>Each line shows only the proportional quoted amount — no rates.</i>

Quotation settings

- Timeline (weeks) — how long from deposit to delivery — appears on the PDF
- Payment Terms — default: 50% deposit on approval, 50% on completion — editable
- Scope Notes — any additional notes you want the client to see

Downloading the PDF

Click **Download PDF** in the Quotation tab. The PDF is generated instantly and named after the project code — e.g. **BAL-001_quotation.pdf**.

The PDF is ready to send directly to the client. It shows your company name, project reference, valid-until date, all line items with amounts, payment terms and scope.

 Save the quotation first (Save Quotation button) before downloading if you want the settings stored in the database.

Proportional amounts — how they work

Each client line item amount is calculated as: **(raw cost ÷ your direct total) × quoted price**. This means the total of all client line items always equals the quoted price exactly. Rounding is absorbed in the last line item.

7. Rhino / Grasshopper Integration

Exporting the geometry data

Once the site survey is complete, click **Export to Rhino (.json)** in the sidebar or header. A JSON file is downloaded named after the project code.

This file contains all the dimensional data the Grasshopper scripts need — nothing has to be retyped in Rhino.

JSON file contents

Section	What it contains
geometry	Total dimensions, stringer length, slope angle, step count and rise/going
handrail	Post count, number of panels, height, rail sides, catch profile
profiles	Post, top rail, bottom rail profile strings (e.g. "40x40")
infill	Infill type, sheet thickness, sheet size chosen by optimizer
panel_cnc	Cut slope edge, height, bounding rectangle, interior angle, total panels

Grasshopper scripts — two components

There are two Python scripts for each project type (balcony and staircase). Paste each into a separate **GHPython** component in Grasshopper, then wire them together.

Script	What it does
balcony_reader.py / staircase_reader.py	Reads the JSON file. Connect the json_path input to a Panel containing the file path. Outputs all parameters as individual wires.
balcony_geometry.py / staircase_geometry.py	Receives parameters from the reader. Outputs geometry objects: stringer line, step treads, posts, rails, infill surfaces, catch frame lines, CNC flat panels.

Grasshopper outputs — what you see in Rhino

Output	Description
stringer_line	The main diagonal line from floor to top landing
step_lines	All tread and riser lines for visual reference
post_lines	Vertical handrail post lines at each section boundary
top_rail_line / bottom_rail_line	Polylines along the top and bottom of the handrail
infill_surfaces	The inclined panel surfaces between posts (3D, in-place)
catch_frame_lines	The 4 inner catch frame bars per panel — inset from posts and rails
cnc_panels	Flat parallelogram surfaces laid out for CNC — all identical
cnc_bounding_rects	Bounding rectangles showing the sheet cut area per panel
report	Text output with full material schedule, cut dimensions and sheet fit check

 Use the reload toggle (Button component) in Grasshopper to force a re-read of the JSON file whenever you update the survey data in the system and export a new file.

Workflow in practice

1. Complete the site survey in the system.
2. Click Export to Rhino — save the JSON into your project folder.
3. Open Grasshopper — connect the reader script to the JSON path.
4. Wire reader outputs to the geometry script inputs.
5. The full 3D railing appears in Rhino. Adjust the origin point to place it correctly in your model.
6. Use the report text output to confirm all dimensions match your quotation.
7. Send cnc_panels to the laser cutter nesting software.

Quick reference — default values

Setting	Default	When to change
Welding consumables	7%	Increase for complex weld patterns or premium wire
Powder coat rate	8,000 RWF/m ²	Get a current quote from your coating supplier
Paint rate	5,000 RWF/m ²	Adjust for primer + topcoat combinations
Contingency	10%	Raise to 15% for complex or uncertain jobs
Business margin	30%	Review quarterly against your actual overhead costs
Quotation validity	30 days	Shorten if steel prices are volatile
Payment terms	50% / 50%	Standard for Rwanda market — adjust per client